

Lab Practical #04:

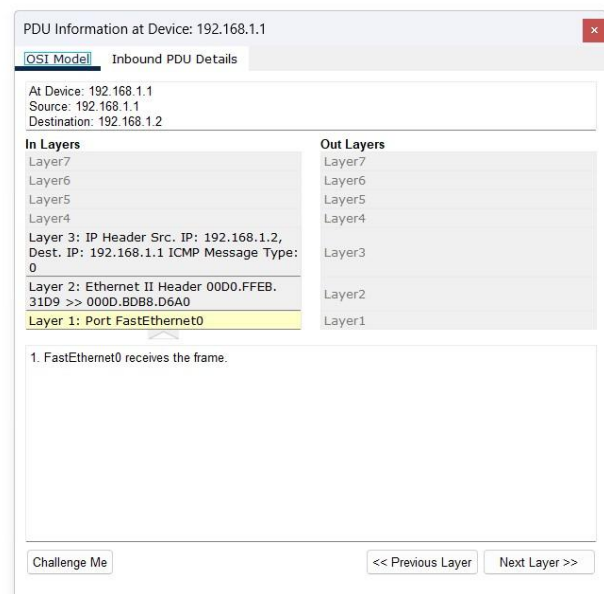
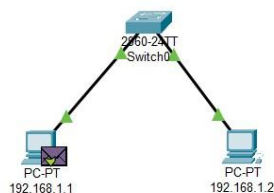
Installation of Network Simulator (Packet Tracer) and Implement different LAN topologies. **Practical Assignment #04:**

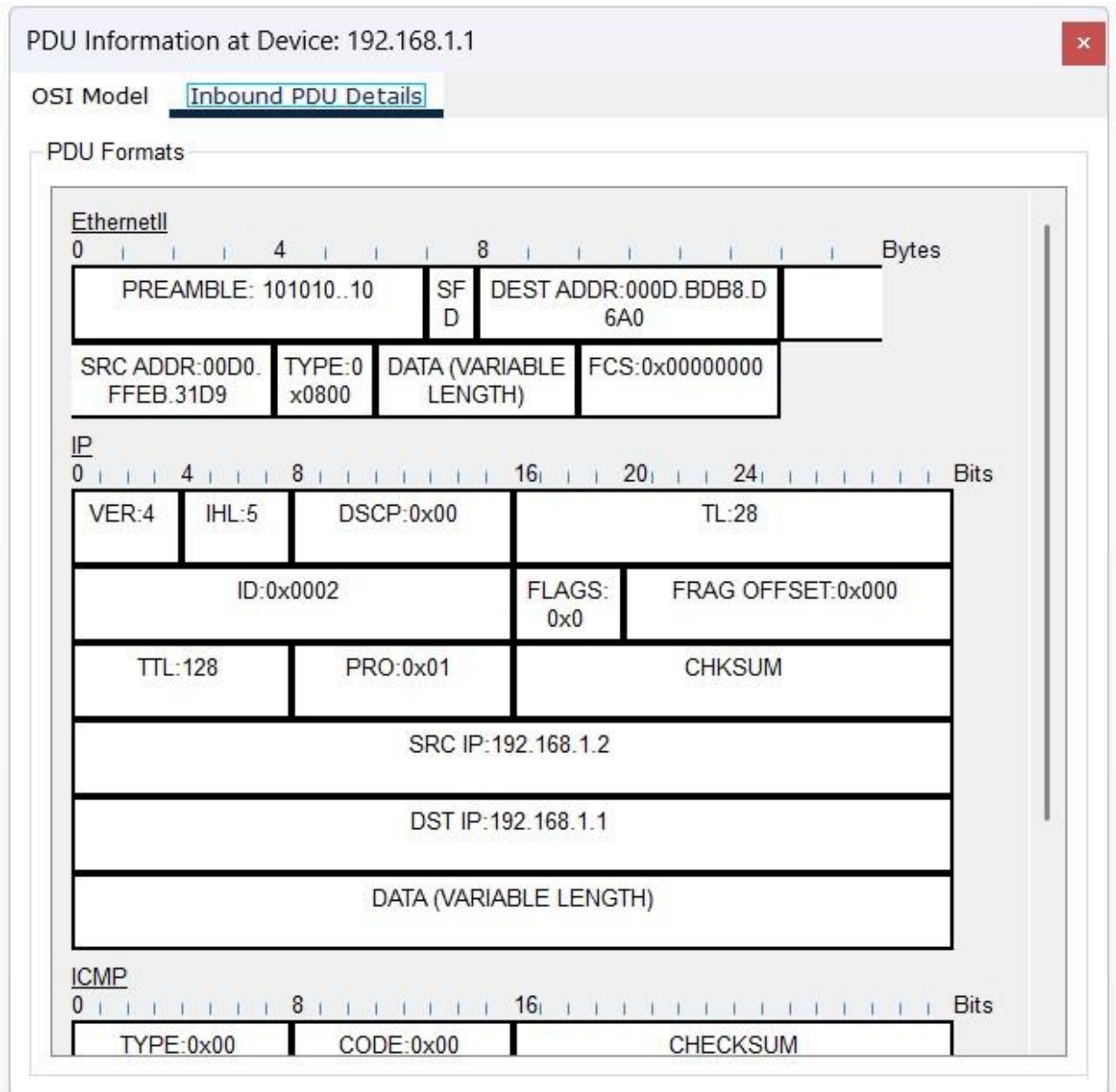
1. Create a simple network with switch and two or more pc. Also check connectivity between them using ping command or PDU utility.
2. Implement different topologies in packet tracer.
 - a. Bus
 - b. Ring
 - c. Star
 - d. Mesh
 - e. Tree

Instructions:

1. Topology-wise Screenshot.
2. Mention IP address of each pc as label.
3. Ping command or PDU screenshot between two pcs.

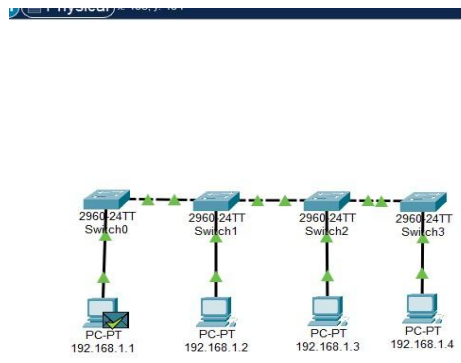
1. Simple Network





2. Topologies

a. Bus



PDU Information at Device: 192.168.1.1

OSI Model Inbound PDU Details

At Device: 192.168.1.1
Source: 192.168.1.1
Destination: 192.168.1.4

In Layers

Layer7
Layer6
Layer5
Layer4
Layer3: IP Header Src. IP: 192.168.1.4, Dest. IP: 192.168.1.1 ICMP Message Type: 0
Layer2: Ethernet II Header 0001.96C0.B87A >> 0001.63D4.2751
Layer 1: Port FastEthernet0

Out Layers

Layer7
Layer6
Layer5
Layer4
Layer3
Layer2
Layer1

1. FastEthernet0 receives the frame.

Challenge Me
<< Previous Layer
Next Layer >>

PDU Information at Device: 192.168.1.1

OSI Model Inbound PDU Details

PDU Formats

EthernetII

PREAMBLE: 101010...10		SF D	DEST ADDR: 0001.63D4.2751
SRC ADDR: 0001.96C0.B87A	TYPE: 0x0800	DATA (VARIABLE LENGTH)	FCS: 0x00000000

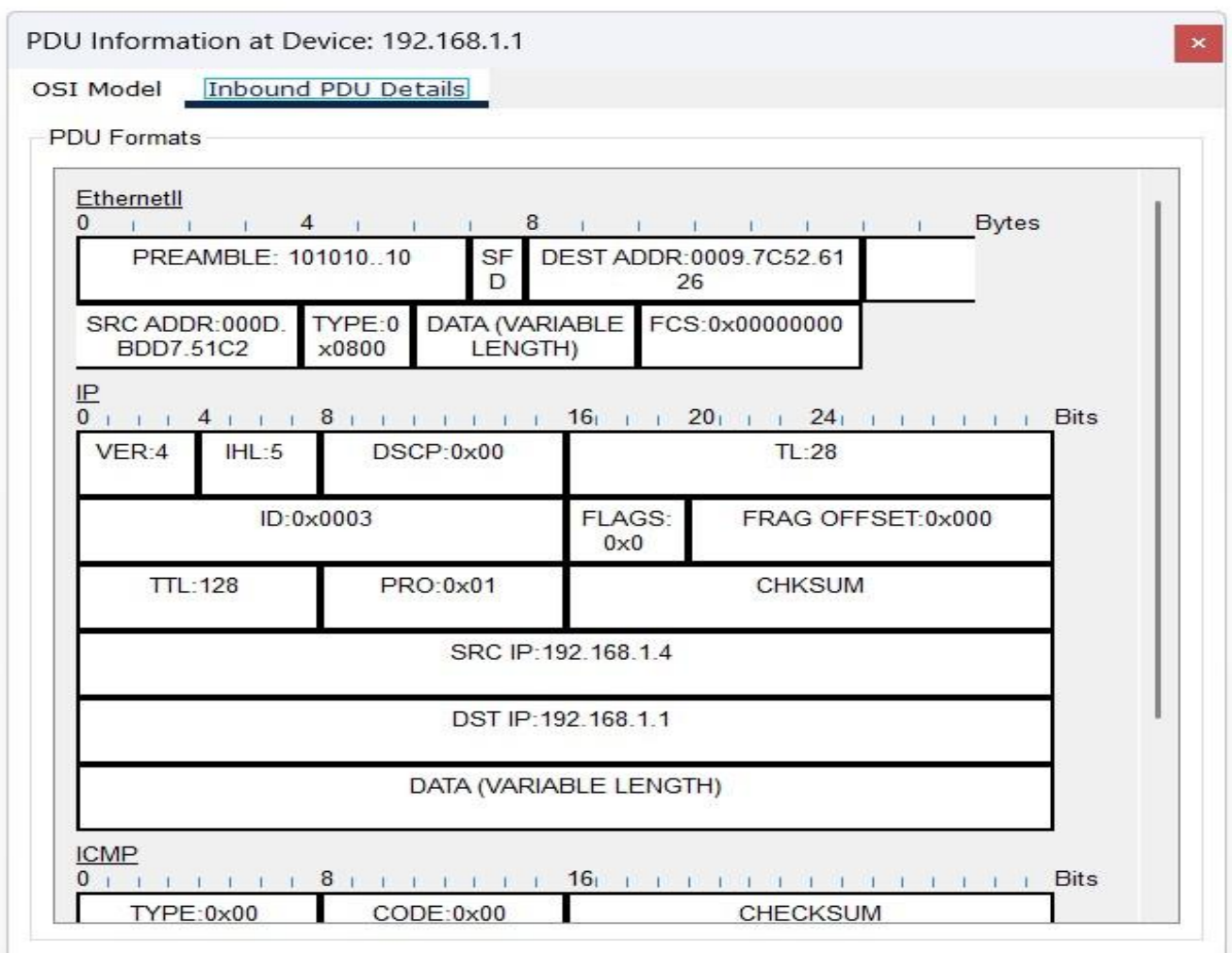
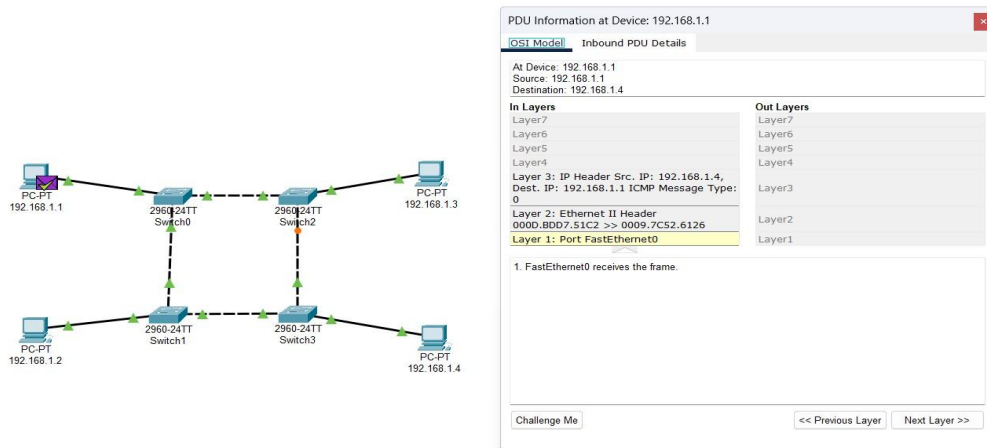
IP

VER: 4	IHL: 5	DSCP: 0x00	TL: 28
ID: 0x0003		FLAGS: 0x0	FRAG OFFSET: 0x000
TTL: 128	PRO: 0x01	CHKSUM	
SRC IP: 192.168.1.4			
DST IP: 192.168.1.1			
DATA (VARIABLE LENGTH)			

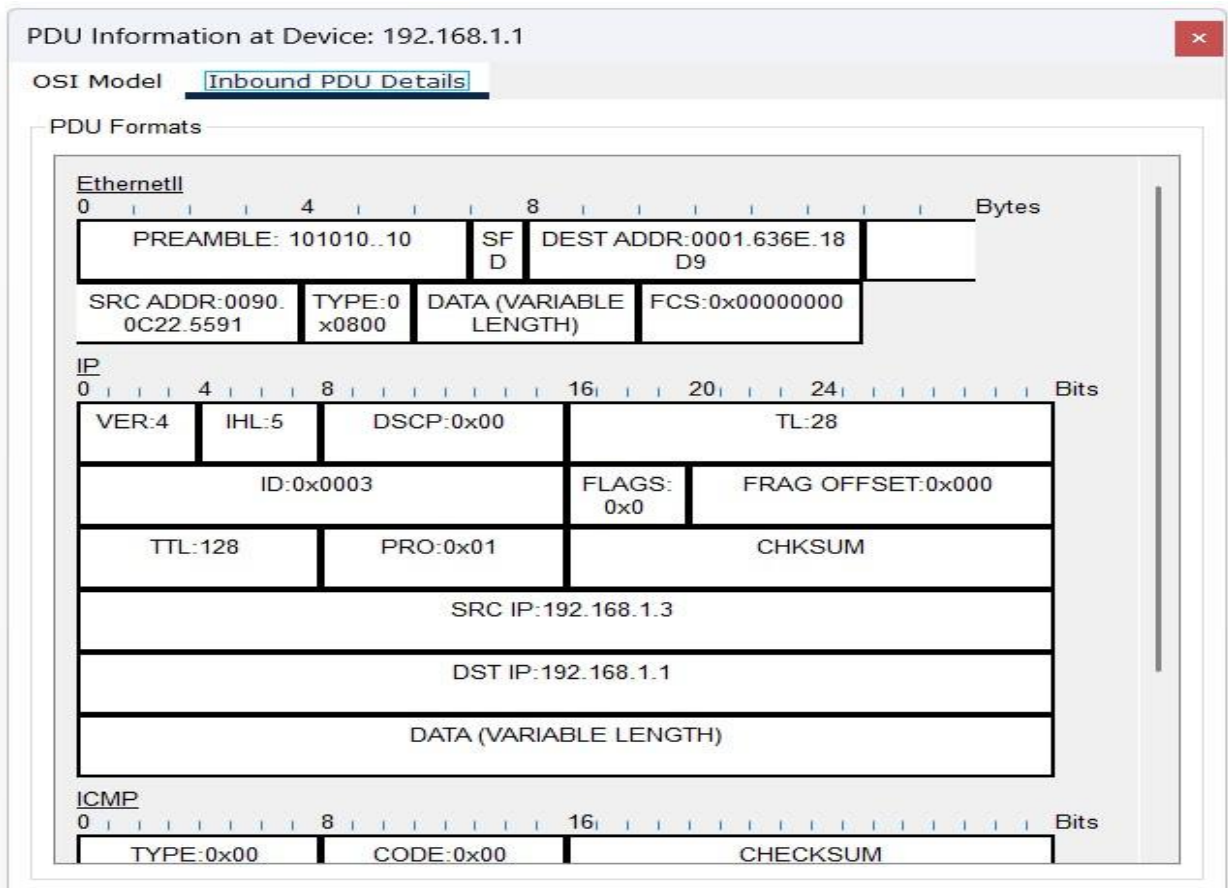
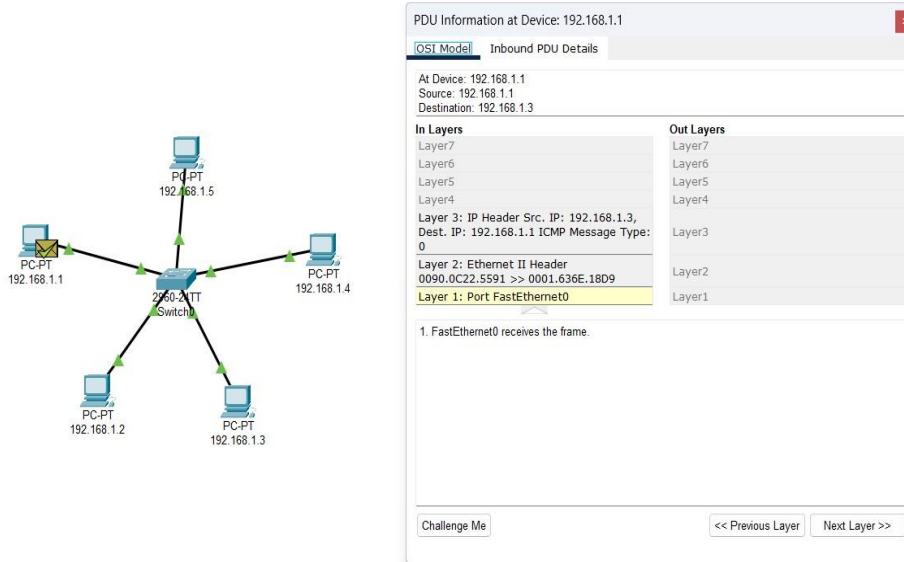
ICMP

TYPE: 0x00	CODE: 0x00	CHECKSUM
------------	------------	----------

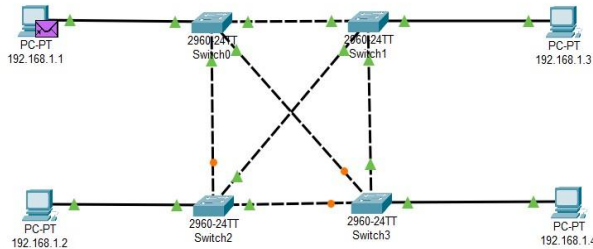
b. Ring



c. Star



d. Mesh



PDU Information at Device: 192.168.1.1

OSI Model Inbound PDU Details

At Device: 192.168.1.1
Source: 192.168.1.1
Destination: 192.168.1.4

In Layers

- Layer7
- Layer6
- Layer5
- Layer4
- Layer3: IP Header Src. IP: 192.168.1.4, Dest. IP: 192.168.1.1 ICMP Message Type: 0
- Layer2: Ethernet II Header 0010.113D.E7E8 >> 00D0.BA3C.4D3C
- Layer1: Port FastEthernet0

Out Layers

- Layer7
- Layer6
- Layer5
- Layer4
- Layer3
- Layer2
- Layer1

1. FastEthernet0 receives the frame.

Challenge Me << Previous Layer Next Layer >>

PDU Information at Device: 192.168.1.1

OSI Model Inbound PDU Details

PDU Formats

EthernetII

0				4				8				Bytes					
PREAMBLE: 101010...10								SF D		DEST ADDR: 00D0.BA3C.4D3C							
SRC ADDR: 0010.113D.E7E8				TYPE: 0x0800				DATA (VARIABLE LENGTH)				FCS: 0x00000000					

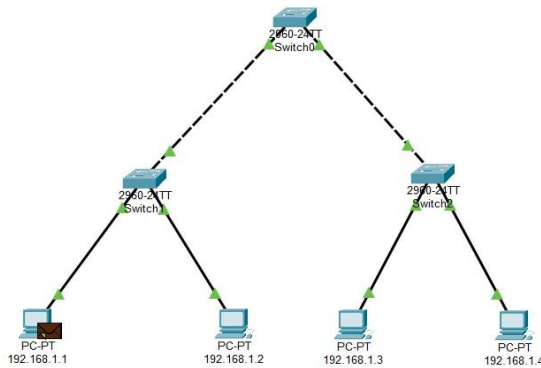
IP

0		4		8		16		20		24		Bits	
VER: 4		IHL: 5		DSCP: 0x00		TL: 28							
ID: 0x0002						FLAGS: 0x0		FRAG OFFSET: 0x000					
TTL: 128				PRO: 0x01				CHKSUM					
SRC IP: 192.168.1.4													
DST IP: 192.168.1.1													
DATA (VARIABLE LENGTH)													

ICMP

0		8		16		Bits	
TYPE: 0x00		CODE: 0x00		CHECKSUM			

e. Tree



PDU Information at Device: 192.168.1.1

OSI Model Inbound PDU Details

At Device: 192.168.1.1
Source: 192.168.1.1
Destination: 192.168.1.3

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer3: IP Header Src. IP: 192.168.1.3, Dest. IP: 192.168.1.1 ICMP Message Type: 0	Layer3
Layer2: Ethernet II Header 0060.70AC.A3BE >> 00D0.9746.2044	Layer2
Layer1: Port FastEthernet0	Layer1

1. FastEthernet0 receives the frame.

Challenge Me << Previous Layer Next Layer >>

PDU Information at Device: 192.168.1.1

OSI Model Inbound PDU Details

PDU Formats

EthernetII

Bytes			
PREAMBLE: 101010...10	SF D	DEST ADDR: 00D0.9746.2044	
SRC ADDR: 0060.70AC.A3BE	TYPE: 0x0800	DATA (VARIABLE LENGTH)	FCS: 0x00000000

IP

Bits			
VER: 4	IHL: 5	DSCP: 0x00	TL: 28
ID: 0x0002		FLAGS: 0x0	FRAG OFFSET: 0x000
TTL: 128	PRO: 0x01	CHKSUM	
SRC IP: 192.168.1.3			
DST IP: 192.168.1.1			
DATA (VARIABLE LENGTH)			

ICMP

Bits		
TYPE: 0x00	CODE: 0x00	CHECKSUM